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Anaesthetics Department
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PRE-OPERATIVE CARE

Pre-Operative Investigations

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Key Points

- Full Guidance: NICE guidance: https://www.nice.org.uk/guidance/NG45

Full Blood Count (FBC)

Age	(FBC should not be requested on the grounds of age alone)
Operation Severity	Intermediate or Major (see Appendix)
ASA	III or IV
Specific Condition	Any surgery requiring G&S or XM (see Appendix) Anaemia suspected Known haematological disorder History of alcohol abuse Dyspnoea (Grade 2 or more) Hypertension Medications: Anti-epileptics

Urea & Electrolytes (U&Es)

Age	U&Es should not be requested on the grounds of age alone
Operation Severity	Intermediate or Major
ASA	III or IV
Specific Condition	Urological Surgery Known CKD3 or more Diabetes or Hypertension Diabetes or hypertension Diuretics/ACE/ARA's or any drug known to cause renal impairment **N/B:** ***Add bicarbonate in severe lung disease & OSA***

Liver Function Tests (LFTs)

Age	Liver function should not be requested on the grounds of age alone
Operation Severity	Biliary Major intra-abdominal
Specific Condition	History of liver disease History of bleeding tendency Previous alcohol excess >30u /week women or >40u/week men Current alcohol excess Super morbid obesity or anorexia Certain drug/co-morbidity combinations (see below)*

*Consider LFTS if one of the risk factors below **and >1** of the drugs listed

Risk factors

- Age greater than 70
- Pre-existing liver disease
- Genetic pre disposition such as alpha-1 antitrypsin deficiency
- Obesity
- Diabetes
- Renal Failure
- Malnutrition
- HIV
- Polypharmacy

Drugs: ACE inhibitors/ARAs, Allopurinol, Anti-diabetic drugs, Aspirin, Augmentin, Azathioprine, Carbamazepine, Carbimazole, Chlorpromazine, Cocaine, Cyclosporin, Ecstasy, Erythromycin, Flucloxacillin, Methotrexate, NSAIDs, Oral Contraceptive, Phenytoin, Ranitidine, Tamoxifen, Tricyclic anti-depressants, Warfarin.

Clotting Tests

Age	A clotting screen should not be requested on the grounds of age alone
Operation	Biliary Open AAA surgery or major vascular reconstructions
Specific Condition	History of liver disease History of bleeding tendency Previous alcohol excess >30u /week women or >40u week man Current alcohol excess Malnutrition Heart Failure NYHA III/IV

Blood Glucose; Random [≥11.1mmol/L diagnostic] or Fasting [≥7.0 mmol/L diagnostic]

- Known impairment of glucose tolerance
- Previous history of Gestational Diabetes
- Diabetic symptoms (e.g. polyuria, polydipsia & unexplained weight loss)
- Age > 40 (>25 if Asian or African Caribbean) **and either of:**
    - Obesity BMI >30
    - Family history of diabetes in first-degree relative.
- Ischaemic heart disease, claudication
- Hypertension
- Abscess, soft tissue or fungal infections, history of recurrent sepsis
- Severe mental health problems

HBA1C [48mmol/mol diagnostic]

Anyone with known diabetes mellitus or a BMI >35

Pregnancy Test

Please see health Board Policy for all women of reproductive age. Consent must be obtained.

Sickle Cell

- Do not routinely offer testing for sickle cell disease or sickle cell trait before surgery.
- Ask the patient if they (or any member of their family) have sickle cell disease.
- Known sickle cell disease: liaise with haematology team before surgery.

Pulmonary Function Tests

- Grade 3 or 4 dyspnoea, if not done in last 6 months, or if acute change in symptoms.
- If PEFR < 70% predicted.
- New or unexplained breathlessness
- Significant neurological or neuromuscular disease, especially if they are immobile
- Oxygen saturation is less than 93% on air

C-Spine

Request **C Spine views in flexion and extension** for:

- Rheumatoid arthritis if not done in last 12 months
- Down's Syndrome if cooperative and not in last 12 months
- Known or suspected atlanto-occipital instability. E.g. severely restricted movement or pain and parasthesia in arms on neck movement

Request **AP and lateral views of C spine** for:

- anaesthetic or surgical assessment in patients with malignant disease with neck pain or with bony metastases

CXR

- There are very few indications for a routine pre-operative chest X-Ray.
- Please discuss with an anaesthetist if you feel a patient warrants one for cardio-respiratory problems, or for missed cancer screening.

ECG

Age	>65, >55 in a smoker
Operation Severity	Intermediate or major
ASA	II, III, or IV
Specific Condition	Hypertension Known or suspected heart disease Known respiratory disease Diabetes Istory of alcohol abuse BMI >35 Detected heart murmur Tachycardia >100 Palpitations Irregular pulse New or unexplained breathlessness

Echocardiography

- First check for previous studies. If performed, ascertain the date and obtain a copy in the patient notes.
- For minor surgery an echocardiogram is not indicated.

Indicated

Heart Failure

1) Documented ischaemic heart disease with reduced functional capacity (<4 METS).
2) Unexplained shortness of breath in the absence of clinical signs of heart failure if ECG and/or CXR abnormal.
3) Signs or symptoms of heart failure.2 These include unexplained shortness of breath, orthopnoea, paroxysmal nocturnal dyspnoea (PND), peripheral oedema.

Murmurs and Valvular Heart Disease

4) Murmur in the presence of cardiac or respiratory symptoms.
5) Murmur in an asymptomatic individual in whom clinical features or other investigation suggest severe structural heart disease.
6) Known valvular heart disease where follow-up echo has been missed (see figure 1, below).

Pulmonary Disease

7) Lung disease with clinical evidence of Cor Pulmonale, and no previous echo.
8) Suspected or established pulmonary hypertension.

Other Indications

9) Proven Arrhythmia (including AF) and no previous echo.
10) Newly diagnosed hypertension with symptoms/findings of LV heart failure and no prev echo.
11) Newly diagnosed LVH and no previous echo.

Not Indicated

1) Minor Surgery.
2) Repeat assessment of previous echocardiogram with no intervening change in clinical status, except in setting of valvular heart disease (see '6.' Above).
3) Routine pre-operative echocardiography.
4) New murmur in absence of cardiac symptoms (including including breathlessness, pre-syncope, syncope or chest pain 2 or orthopnoea/PND/Bilateral pitting peripheral oedema) or ECG changes.

Figure 1. BSE Recommended Follow-up Echo Intervals, Valvular Heart Disease

Valve Lesion	Severity	Recommended Interval
Aortic Stenosis	Severe	6-12 months
	Moderate	1-2 years (1 year if Vmax3.5-3.9m/s)
	Mild	3-5 years
Aortic Regurgitation	Severe	6-12 months
	Moderate	1-2 years
	Mild	not required[1]
Mitral Stenosis	Severe	6-12 months
	Moderate	1-2 years
	Mild	3-5 years
Mitral Regurgitation	Severe	6-12 months
	Moderate	1-2 years
	Mild	not required[1]
Tricuspid/Pulmonary Regurgitation	Severe	1 year
	Mild/Moderate if abnormal valve or RV	2 years if abnormal valve or RV
	Mild/moderate no abnormal valve or RV	Not required
Pulmonary Stenosis	Severe	1 year
	Moderate	2 years
	Mild	3-5 years
Mechanical Valves	Normal baseline post procedure echo	not required
	Regurgitation on baseline echo	by native valve criteria
Bioprosthetic Valves	AV/TV/MV <60yrs	Annual after 1st 5 years
	AV/TV/MV >60yrs	Annual after 1st 10 years
TAVI/Valve repairs/Root dilation/	Discuss with Dr Royle	

Sleep Studies

See *YG CG 310 Obstructive Sleep Apnoea*

References

- https://www.nice.org.uk/guidance/ng45/evidence/full-guideline-87258149468
- Indications for echocardiography.pdf (bsecho.org)

Appendix 1: NICE Surgical Grading

Recommendations for specific surgery grades (minor, intermediate, and major or complex) and ASA grades

The following recommendations are specific to surgery grade and ASA grade.

Surgery grades

Surgery grades	Examples
Minor	• excising skin lesion • draining breast abscess
Intermediate	• primary repair of inguinal hernia • excising varicose veins in the leg • tonsillectomy or adenotonsillectomy • knee arthroscopy
Major or complex	• total abdominal hysterectomy • endoscopic resection of prostate • lumbar discectomy • thyroidectomy • total joint replacement • lung operations • colonic resection • radical neck dissection

Appendix 2: Operations requiring Group & Save and cross matching

Speciality	Procedure	
Gynaecology	Abruption	G/S
	Anterior Rep (GYN)	G/S
	Caesarian Section	G/S
	Colposuspension	G/S
	Concealed Pregnancy	G/S
	CONE BIOPSY	G/S
	Delorme Procedure	G/S
	Ectopic Pregnancy	G/S
	Evacuation, Retained Products	G/S
	External Cepahlic Version	G/S
	Hysterectomy, Total Abdominal TAH)	G/S
	Hysterectomy, Vaginal	G/S
	Hysterectomy, Wertheims	4
	Oophrectomy	G/S
	Ovarian Cyst	G/S
	Pelvic Floor Repair	G/S
	Retained Placenta	G/S
	Reversal of sterilization	G/S
	Ruptured Uterus	6
	Termination O P	G/S
	Trial of Scar	G/S
Orthopaedics	Austin Moore Prosthesis	G/S
	Dynamic Hip Screw	G/S
	Elbow Replacement	G/S
	Hemiarthroplasty	G/S
	IM Nailing (unless polytrauma)	G/S
	Removal of JM Nail	G/S
	THR, Bilateral	G/S
	THR, Revision	4
	THR, Total Hip Replacement	G/S
	TKR, Bilateral	G/S
	TKR, Bilateral, with autologous support	G/S
	TKR, Revision	G/S
	TKR, Primary with autologous support	G/S
	TKR, Primary Total Knee Replacement	G/S
General Surg	A P Resection	2
	Amputation, Above Knee	G/S
	Ampution, Below Knee	2
	AP Excision	2
	(Lower) Anterior Resection rectum	2
	Appendix	G/S
	Biopsy	G/S
	Biopsy, Renal	G/S
	Block dissection neck	2
	Cholecystectomy	G/S
	Choledocho-jejunostomy	G/S
	Colectomy	G/S
	Colonoscopy	G/S
	Colostomy	G/S
	Endoscopy	G/S
	ERCP (GI Endoscopy)	G/S

	Excision Thyroglossal Cyst	G/S
	Fundoplication (Abd)	G/S
	Gastroentrostomy	G/S
	Gastro-jejunostomy	G/S
	Hartmans Procedure	2
	Hemicolectomy	G/S
	Hernia Repair, Incisional	G/S
	Ileostomy	G/S
	Ileostomy, reversal	G/S
	Insertion Tenchkoff Catheter	G/S
	Laparoscopy	G/S
	Laparotomy	2
	Mastectomy	G/S
	Mastectomy, Radical	2
	Panproctocolectomy	2
	Pelvic Clearance	5
	Perforated Ulcer	G/S
	Pouch procedure	2
	Proctectomy	G/S
	Proctocolectomy	2
	Sigmoid Colectomy	G/S
	Small Bowel Resection	G/S
	Splenectomy	G/S
	Thyroidectomy, Partial	G/S
	Thyroidectomy, Sub-total	G/S
	Thyroidectomy, Total	G/S
	Tracheostomy	2
	Vagotomy	G/S
	Wide local excision	G/S
Urology	A-V Shunt (Urology)	G/S
	Cystectomy	3
	Cystoscopy	G/S
	Nephrectomy	G/S
	Nephrolithotomy	G/S
	PCNL	G/S
	Prostatectomy, Open	G/S
Vascular	By-pass, Femoral-popliteal	G/S
V	Aneurysm, Abdominal Aortic repair	10
	Aneurysm, Elective Repair	4
	Aneurysm, Femoral	3
	Axillo-bifomoral graft	2
	By-pass, Aorto-Bifemoral	2
	By-pass, Aortomesenteric	4
V	By-pass, Femoro-Tibial	G/S
V	By-pass, Ileo-femoral	G/S
V	Carotid Endarterectomy	G/S
V	Cervical Sympathectomy-Endoscopic	G/S
V	Embolectomy	G/S
V	Femoro-femoral X-over graft	G/S